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(vii)	The numerical aperture of a fiber with a refractive index of the core and cladding 1.54 and 1.50 respectively is a) 0.349 b) 34.9 c) 349 d) 3.49	1	2
(viii)	In a diffraction pattern due to a single slit of width a , the first minimum is observed at an angle 30° when light of wavelength $5000\text{\AA}$ is incident on the slit. The first secondary maximum is observed at an angle of a) $\sin^{-1}(2/3)$ b) $\sin^{-1}(1/2)$ c) $\sin^{-1}(3/4)$ d) $\sin^{-1}(1/4)$	1	3
(ix)	Missing orders are found in case of double slit diffraction patterns due to a) unequal values of two slit widths b) superposition of diffraction minima and interference maxima c) superposition of diffraction maxima and interference minima d) oblique incidence of light	1	3
(x)	Relaxation time for damped electrical oscillation is given as a) R/L b) $2R/L$ c) $2L/R$ d) $LR/2$	1	1
(xi)	An electron has a speed of 500m/s with an accuracy of 0.005% , the uncertainty in the position of the electron will be a) 4.62mm b) 46.2cm c) 4.62 m d) 46.2 mm	1	3
(xii)	The effective number of atoms per unit cell of an B.C.C. lattice is a) 4 b) 6 c) 14 d) 2	1	4

GROUP – B
(Short Answer Type Questions)
(Answer any three of the following) $3 \times 5 = 15$

SL	Question	Marks	Co
2.	(i) Calculate the ratio of the spontaneous emission to the stimulated emission at a temperature 300K for the sodium D line.	3	2
	(ii) Define quality factor.	2	1
3.	(i) Calculate the de-Broglie wavelength associated with the proton moving velocity 10^7m/s .	2	5
	(ii) Light of frequency $6 \times 10^{14}\text{Hz}$ falls on a metal surface having work function 1.9eV . Find maximum velocity of	3	5

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surface electrons.

4.	(i)	Light of a wavelength $2000\text{\AA}$ falls on an aluminium surface having work function 4.2eV . Find out the threshold wavelength and stopping potential.	2	2
	(ii)	Obtain the Packing fraction of BCC and FCC structures.	3	4
5.	(i)	Why electron can't exist in the nucleus, explain.	2	5
	(ii)	For one dimensional motion of a particle the normalized wave function is $\psi = \sqrt{\beta} \exp(-\beta x)$. Calculate the probability of finding the particle in the region between $x = -2$ to $x = 2$.	3	5
6.	(i)	A damped vibrator starting from rest has initial amplitude 0.5 m . It reduces to 0.4 m after 50 complete cycles, each of period 2 s . Find logarithmic decrement.	3	1
	(ii)	Explain sharpness of resonance using graph.	2	1

GROUP – C

(Long Answer Type Questions)

(Answer any three of the following) $3 \times 15 = 45$

SL	Question	Marks	Co
7.	(i) Interrelate Einstein's A and B coefficients.	5	2
	(ii) A double slit of 0.5 mm separation is illuminated by light of blue cadmium light of wavelength $5000\text{\AA}$. How far behind the slits must go to obtain fringes that are 0.1 cm apart? The wavelength of the blue cadmium light is $4800\text{\AA}$.	2	3
	(iii) Describe the working principle of optical fibre. Also give two uses of optical fibre.	2+1	2
	(iv) A particle of mass 0.01 kg is subjected to a restoring force of 0.5 N/m , a damping force 10^{-4} Kg/s and an external sinusoidal restoring force of constant amplitude. For, what frequency of the driving force is velocity resonance achieved? Calculate the Q-factor and bandwidth at resonance.	5	1
8.	(i) Obtain an expression for intensity of double slit Fraunhofer diffraction pattern.	5	3
	(ii) Sodium light of wavelengths 589 nm and 589.6 nm are made incident normally on a grating having 700	5	3

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lines/mm. Calculate the angular dispersion of these lines in the spectrum of first order.

	(iii)	Explain Nicol as polarizer as well as analyzer.	5	2
9.	(i)	Newton's rings are observed in reflected light of wavelength $5900\text{\AA}$. The diameter of 10 th dark ring is 0.5cm. Find the radius of the curvature of the lens and the thickness of the air film.	5	3
	(ii)	Obtain a relation for spacing between lattice planes.	5	4
	(iii)	Using Brewster's law, prove that the angle between reflected ray and refracted ray is 90° .	3	2
	(iv)	Light of frequency $6 \times 10^{14}\text{Hz}$ falls on a metal surface having work function 1.9eV. Find maximum velocity of surface electrons.	2	5
10.	(i)	Show that intensities of the first and second secondary maxima are approximately 4.5% of I_0 and 1.62% of I_0 respectively, where I_0 have usual meaning.	3	3
	(ii)	If the wavelength of the radiation of the He-Ne laser is $6328\text{\AA}$, calculate the energy of the radiation.	2	3
	(iii)	Discuss the construction and working of He-Ne laser.	5	2
	(iv)	Find the probability of finding a particle lying in the region $1/b < x < 3/b$, where $\psi(x) = Ae^{-bx}$ for $0 < x < \alpha$ and $\psi(x) = 0$ for $\alpha < x < 0$.	3	5
	(v)	Distinguish between Fresnel and Fraunhofer class of diffraction.	2	3
11.	(i)	Set up the differential equation of forced harmonic motion and solve it.	2+3	3
	(ii)	In a crystal, a lattice plane cuts intercepts of 3a, 2b and 6c along the three axes, where a, b and c are primitive vectors of the unit cell. Determine the miller indices of the given plane.	3	4
	(iii)	Using the expression of phase velocity, prove that the rest mass of the photon is zero. Also determine its momentum.	5	5
	(iv)	The density of BCC iron is 7900 Kg/m^3 and its atomic weight is 56. Calculate the side of the unit cell.	2	4